

Artin-Schreier Configurations and Higher Dimensional Geproci Sets

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Definition

A finite set Z in $\mathbb{P}_F^n$ is **geproci** if the projection $\overline{Z}$ of Z from a general point to a hyperplane $H = \mathbb{P}_F^{n-1}$ is a complete intersection in H .

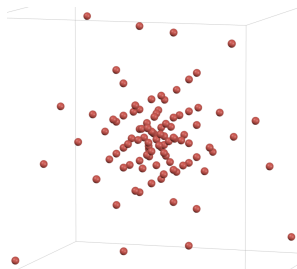
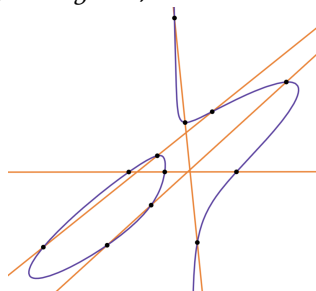
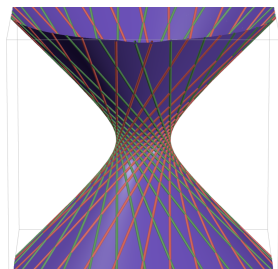
Geproci stands for **g**eneral **p**rojection is a **c**omplete **i**ntersection.

In characteristic 0, the only nontrivial examples known are for $n = 3$. In this case a hyperplane is a plane. A reduced set of points in a plane is a complete intersection if it is the transverse intersection of two algebraic curves.

For $\#Z = ab$ ($a \leq b$), Z is (a, b) -geproci if $\overline{Z}$ is the intersection of a degree a curve and a degree b curve.

Species of Geproci

Grids, half-grids, and non-half-grids



Definition

Given a $k \geq 0$ and field F , a **k -spread** of $\mathbb{P}_F^n$ is a set of mutually-skew k -planes that partition $\mathbb{P}_F^n$.

Spreads are known to exist for $F =$ any finite field if and only if $k + 1$ divides $n + 1$, and also for $k + 1 = (n + 1)/2$ for $F = \mathbb{R}$.

Spreads are instrumental for the proof that $\mathbb{P}_{\mathbb{F}_q}^3$ is a $(q + 1, q^2 + 1)$ -geproci half-grid in $\mathbb{P}_{\mathbb{F}_q}^3$ [K– 24]. In this case, a spread is a partition of $\mathbb{P}_{\mathbb{F}_q}^3$ into lines.

It is possible to construct a spread of k -dimensional hyperplanes inside $\mathbb{P}_{\mathbb{F}_q}^n$ if and only if $k + 1$ divides $n + 1$.

Let $p(x) \in \mathbb{F}_q[x]$ be irreducible, monic, and degree $k + 1$, and let P be its companion matrix. Then one can construct a spread of spaces of the form

$$\text{row} \left(\underbrace{0 \quad \cdots \quad 0 \quad \boxed{I_{k+1}} \quad A_1 \quad \cdots \quad A_j}_{\frac{n+1}{k+1}} \right) : A_i \in \mathbb{F}_q[P].$$

Maximal Partial Spreads

Note that a spread over $\mathbb{P}_{\mathbb{F}_q}^3$ comprises $q^2 + 1$ mutually-skew lines.

Definition

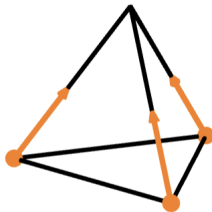
A **partial spread** of $\mathbb{P}_{\mathbb{F}_q}^3$ with **deficiency** d is a set of $q^2 + 1 - d$ mutually-skew lines. A **maximal partial spread** is a partial spread of positive deficiency that is not contained in any larger partial spread.

Maximal partial spreads are also instrumental to the study of geproci sets; in fact, given a maximal partial spread $\mathcal{M}$, the set $\mathbb{P}_{\mathbb{F}_q}^3 \setminus (\bigcup_{L \in \mathcal{M}} L)$ is geproci [K– 24].

Non-Reduced Geproci Schemes

We can construct a flat family of zero-dimensional schemes in $\mathbb{P}^n$ whose general fiber is reduced and not geproci, while the special fiber is nonreduced and geproci. Thus the complete-intersection property of a general projection is **created** at the special fiber.

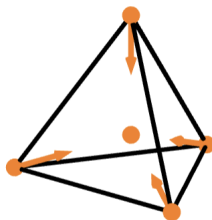
Take $I = (zw, yw, yz, x^2)$ over a field of characteristic 2. Then I is the ideal of a length-6 0-scheme whose general projection is a $(2, 3)$ -complete intersection. No $(2, 3)$ half-grid is known in characteristic 0.



Another Example

Let $Z = \{(1, 0, 0, 0)^2, (0, 1, 0, 0)^2, (0, 0, 1, 0)^2, (0, 0, 0, 1)^2, (1, 1, 1, 1)\}$ in characteristic 2, with each infinitely-near point corresponding to the line containing $(1, 1, 1, 1)$. Then Z is $(3, 3)$ -geproci. It is a non-trivial non-half-grid.

No nontrivial $(3, 3)$ -geproci sets are known in characteristic 0.



The general projection of Z is the base locus of a quasi-elliptic fibration, which only exist in characteristics 2 & 3.

Characteristic 0 Behavior

Consider the 60 projective points $Z \subseteq \mathbb{P}_{\mathbb{C}}^3$ given by the 120 rays of the H_4 root system. Then Z is $(6, 10)$ -geproci.

Furthermore, Z can be partitioned into a union of 12 lines L_i with two common transversals. Defining $P_i = T \cap L_i$, we get the length-60 curvilinear configuration

$$Z' = \bigcup_{i=1}^{12} P_i^5|_{L_i}$$

Notably, Z' is $(5, 12)$ -geproci, but it is the special fiber of a pencil of $(6, 10)$ -geproci configurations.

Artin-Schreier Configurations

Let K be an algebraically closed field, $\text{char } K > 0$, and $\mathbb{F}_q \subseteq K$.

Definition

Let $L_1, \dots, L_r \subseteq \mathbb{P}_K^n$ be distinct lines through a common point O . An $\mathbb{F}_q$ -**Artin-Schreier configuration** supported on $L_1, \dots, L_r$ is a finite set $Z = \bigcup_{i=1}^r B_i$ where $B_i \subseteq L_i \setminus \{O\}$ is an affine copy $\beta_i + \lambda_i \mathbb{F}_q$ for $\beta_i, \lambda_i \in L_i$.

Each B_i is called an **Artin-Schreier orbit**.

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Each B_i is called an **Artin-Schreier orbit**. They are named after the Artin-Schreier polynomials:

$$\prod_{a \in \mathbb{F}_q} (t - a) = t^q - t,$$

and

$$\prod_{a \in \mathbb{F}_q} (t - (\beta + \lambda a)) = (t - \beta)^q - \lambda^{q-1}(t - \beta).$$

Definition

The **evaluation map** on $Y \subseteq \mathbb{P}_K^n$ in **degree** N is

$$H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(N)) \rightarrow K^{\#Y},$$

taking every polynomial of degree N and evaluating it at each point of Y .

Artin-Schreier Extensions

Definition

The **evaluation map** on $Y \subseteq \mathbb{P}_K^n$ in degree N is

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Lemma (4.1 CFFHMSS)

Let $Y \subseteq \mathbb{P}_K^{n-1}$ be a reduced complete intersection of type $(d_1, \dots, d_{n-1})$, let $O \in \mathbb{P}_K^n \setminus \mathbb{P}_K^{n-1}$, and let $Z \subseteq \mathbb{P}_K^n$ be obtained by placing an $\mathbb{F}_q$ -Artin-Schreier orbit on each line of the cone $C_O(Y)$. If the evaluation map of Y in degree $q - 1$ is surjective, then Z is a complete intersection of type $(d_1, \dots, d_{n-1}, q)$.

Geproci-Induction of Artin-Schreier Sets

Theorem (3.4 CFFHMSS)

Let $Z \subseteq \mathbb{P}^3$ be an $\mathbb{F}_q$ -Artin-Schreier configuration supported on r concurrent lines spanning $\mathbb{P}^3$. Then Z is an (r, q) -geproci set if and only if $r \leq q$.

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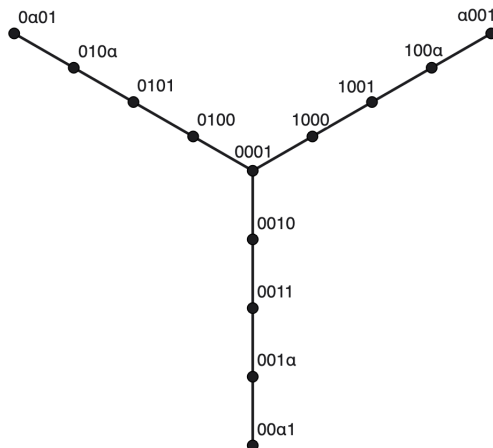
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Theorem (4.2 CFFHMSS)

Let $X \subseteq \mathbb{P}_K^n$ be a geproci set whose general projection to $\mathbb{P}_K^{n-1}$ is a complete intersection of type $(d_1, \dots, d_{n-1})$, and let $Z \subseteq \mathbb{P}_K^{n+1}$ be an $\mathbb{F}_q$ -Artin-Schreier extension. If the general projection X' of X into $\mathbb{P}_K^{n-1}$ admits a surjective evaluation map in degree $q - 1$, then Z is a geproci set whose general projection is a complete intersection of type $(d_1, \dots, d_{n-1}, q)$.

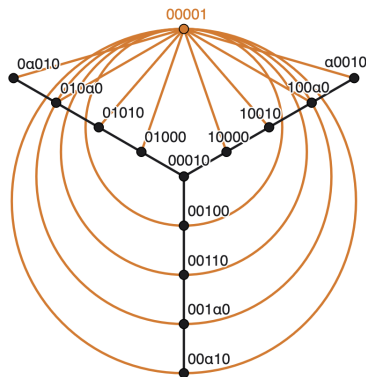
Example in Characteristic 2

Start with a $(3, 4)$ -geproci $\mathbb{F}_4$ -Artin-Schreier configuration Z in $\mathbb{P}_K^3$.



Example in Characteristic 2

Take the cone of Z over a point in $\mathbb{P}_K^4 \setminus \mathbb{P}_K^3$. On each of the 12 obtained lines, place an $\mathbb{F}_8$ -Artin-Schreier orbit.



After a general projection to $\mathbb{P}_K^3$, the 96 points form a $(3, 4, 8)$ -complete intersection. This example can be imitated to create geproci sets in any dimension and positive characteristic!

The Return of Non-Reduced Geproci

Let $L_1, \dots, L_4 \subset \mathbb{P}^4$ be four general lines through a common point O , and choose general points $P_i \in L_i \setminus \{O\}$. Set

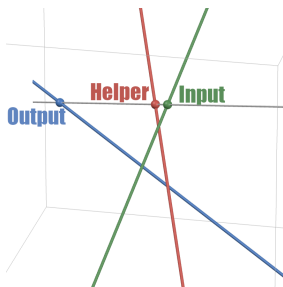
$$Z = \bigcup_{i=1}^4 P_i^2|_{L_i}.$$

The scheme Z has length 8 and is nondegenerate in $\mathbb{P}^4$. Its general projection to $\mathbb{P}^3$ is a complete intersection of three quadrics if and only if $\text{char}(K) = 2$. In particular, in characteristic 2, the scheme Z is $(2, 2, 2)$ -geproci.

Transversal-Mappings between n -planes

Definition

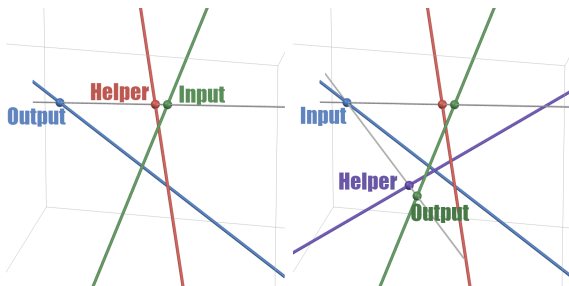
Given three n -planes $H_1, H_2, H_3 \subseteq \mathbb{P}_F^{2n+1}$ where $H_1 \cap H_2 = H_2 \cap H_3 = \emptyset$, we can define the function $\pi(H_1, H_2, H_3) : H_1 \rightarrow H_3$ as follows: take $p \in H_1$. Then there is a unique line T such that $p \in T$, $T \cap H_2 \neq \emptyset$, and $T \cap H_3 \neq \emptyset$. Then define $\pi(H_1, H_2, H_3)(p) = T \cap H_3$. This is the **projection** of H_1 to H_3 via H_2 .



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Definition

A **groupoid** is a category $\mathcal{G}$ where every morphism is invertible.

- For any object $G \in \mathcal{G}$, $\text{Hom}_{\mathcal{G}}(G, G) = \text{Aut}_{\mathcal{G}}(G)$ is a group.

$\text{Aut}_{\mathcal{G}}(G)$ is a “group of the groupoid.”

- Whenever $\text{Hom}_{\mathcal{G}}(G_1, G_2) \neq \emptyset$, then $\text{Aut}_{\mathcal{G}}(G_1) \cong \text{Aut}_{\mathcal{G}}(G_2)$.

So when $\text{Hom}_{\mathcal{G}}(G_1, G_2) \neq \emptyset$ for all $G_1, G_2 \in \mathcal{G}$, $\mathcal{G}$ induces only one group of the groupoid, up to isomorphism. Then it makes sense to say “the” group of the groupoid, $\text{Aut}_{\mathcal{G}}$.

Groupoids of n -planes

Theorem

Let $\mathcal{H}$ be a set of n -planes in $\mathbb{P}_F^{2n+1}$. Define Π to be the composition-closure of the set of functions $\{\pi(H_i, H_j, H_k) \mid H_i, H_j, H_k \in \mathcal{H}, H_i \cap H_j = H_j \cap H_k = \emptyset\}$. Then $(\mathcal{H}, \Pi)$ is a groupoid.

In this case, any group of the groupoid is a subgroup of $\text{Aut}(\mathbb{P}_F^n) \cong \text{PGL}(n+1, F)$.

What can we say about this groupoid and its corresponding group(s)? In characteristic 0, when is it finite versus infinite? What is the relationship (if $\exists$) between $\text{Aut}_{(\mathcal{H}, \Pi)} \leq \text{Aut}(\mathbb{P}^n)$ and $\text{Aut}(\mathcal{H}) \leq \text{Aut}(\mathbb{P}^{2n+1})$? In characteristic 0 and $n = 1$, it is known that when $\text{Aut}_{(\mathcal{H}, \Pi)}$ is finite, then orbits of points under this groupoid are geproci [CFFHMSS 25+].

Ganger's Results

In her 2024 thesis, Ganger used the technique of **transversals** to prove the following theorem:

Theorem (Ganger Corollary 2.5)

The group of the groupoid for the spread induced by the degree-2 field extension $\mathbb{F}_{q^2}/\mathbb{F}_q$ over a finite field is isomorphic to the quotient $\mathbb{F}_{q^2}^/\mathbb{F}_q^* \cong C_{q+1}$.*

Definition

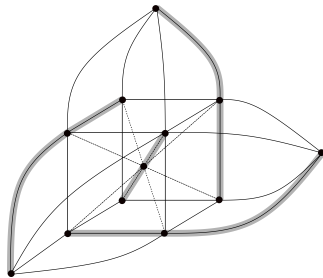
Given a set of lines $\mathcal{L}$ in $\mathbb{P}_F^3$, a **transversal** is a line T in $\mathbb{P}_F^3$ such that $T \cap \bar{L} \neq \emptyset$ for all $L \in \mathcal{L}$.

The Hopf spread has exactly two transversals T_1, T_2 for any finite field. The intersection of transversal with a line $L \in \mathcal{S}$ is a fixed point of $\text{Aut}_{(\mathcal{S}, \Pi)}(L)$!

Example: The D_4 configuration

Theorem (K-)

Let $\mathcal{L}$ be the 16 lines of the $(12_4, 16_3)$ configuration D_4 and let Π be the composition-closure of the projection functions. Then the group of the groupoid $\mathcal{G} = (\mathcal{L}, \Pi)$ is $\text{Aut}_{\mathcal{G}} \cong D_6$.



In fact, one can find a subset of six lines whose group of the groupoid is D_6 [K- 25+]. Such a group is impossible to find in skew configurations [Favacchio 25+].

Finite Groupoids for $n = 1$

The finite subgroups of $\mathrm{PGL}(2, \mathbb{C})$ are C_n , D_{2n} , A_4 , S_4 , and A_5 . All of these except of the dihedral groups can be realized as the groupoid of a configuration of lines in $\mathbb{P}_{\mathbb{C}}^3$ [F 25+].

In characteristic $p > 0$, we can also attain groups $C_p \times C_p$, $C_p \rtimes C_{p-1}$, and D_{2n} for $p \mid n$.

Finite Groupoids for $n = 2$

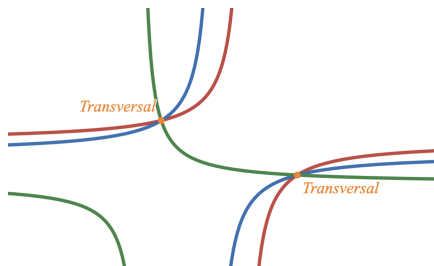
In characteristic 0, the only known finite groups of the groupoid are cyclic. In these configurations, the orbit of a generic point is trivially geproci: the orbit restricted to each plane is a set of collinear points, and the whole orbit only spans a $\mathbb{P}^3$, forming a half-grid.

Blowing Up the n -planes

Consider the n -planes $U, V, W \subseteq \mathbb{P}^{2n+1}$, the blowup map $\beta : \text{Bl}_W(\mathbb{P}^{2n+1}) \rightarrow \mathbb{P}^{2n+1}$, and the exceptional locus

$E = \beta^{-1}(W) \cong \mathbb{P}^n \times \mathbb{P}^n$. Then the projection $\pi : U \xrightarrow{V} W$ extends to a map $\tilde{\pi} : U \rightarrow E$. In the Chow ring $A(E) \cong \mathbb{Z}[\alpha, \beta]/(\alpha^{n+1}, \beta^{n+1})$,

$$[\tilde{\pi}(U)] = \sum_{i=0}^n \alpha^i \beta^{n-i}, \text{ namely, } [\tilde{\pi}(U)]^2 = n + 1.$$



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Three n -planes $U, V, W \subseteq \mathbb{P}^{2n+1}$ uniquely determine a rational normal scroll $\mathbb{P}^1 \times \mathbb{P}^n \cong S \subseteq \mathbb{P}^{2n+1}$ (the Segre embedding, up to coordinates). Then $\tilde{\pi}(U) = \tilde{S} \cap E$.

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Example

Consider $\mathcal{L} = \left\{ (0 \ I_2), (I_2 \ 0), (I_2 \ I_2), \begin{pmatrix} I_2 & \zeta_6 & 1 \\ & 0 & \zeta_6^5 \end{pmatrix}, \begin{pmatrix} I_2 & \zeta_6 & 0 \\ & 1 & \zeta_6^5 \end{pmatrix} \right\}$ in $\mathbb{P}_{\mathbb{C}}^3$. Then $\text{Aut}_{\mathcal{L}} \cong A_4$, and the six induced bidegree-(1,1) curves in E have a stabilizer group $S_4 \rtimes C_2$.

The Conic Invariant

Let $H_0, \dots, H_4 \subseteq \mathbb{P}_F^5$, and let E be the exceptional locus after blowing up H_0 . Then the six pairs of the remaining planes each induce an $\alpha^2 + \alpha\beta + \beta^2$ -surface in E that meet each other in a $(12_3, 6_6)$ -configuration.

Set $A = (M_1 - M_2)^{-1}(M_2 - M_3)$ and $B = (M_1 - M_2)^{-1}(M_2 - M_4)$. Then the six surfaces satisfy the **conic condition** if and only if $\det(AB - BA) = 0$.

When do we get nine points in Cayley-Bacharach position? In Hesse position?